De Broglie Hypothesis

Louis de Broglie

- 1892 – 1987
- French physicist
- Originally studied history

- Was awarded the Nobel Prize in 1929 for his prediction of the wave nature of electrons

Wave Properties of Particles

- Louis de Broglie postulated that because photons have both wave and particle characteristics, perhaps all forms of matter have both properties.
- The **de Broglie wavelength** of a particle is

Frequency of a Particle

- In an analogy with photons, de Broglie postulated that a particle would also have a frequency associated with it

- These equations present the dual nature of matter:
  - Particle nature, p and E
  - Wave nature, λ and f

Complementarity

- The **principle of complementarity** states that the wave and particle models of either matter or radiation complement each other.
- Neither model can be used exclusively to describe matter or radiation adequately.

Davisson-Germer Experiment

- If particles have a wave nature, then under the correct conditions, they should exhibit diffraction effects.
- Davisson and Germer measured the wavelength of electrons.
- This provided experimental confirmation of the matter waves proposed by de Broglie.

Wave Properties of Particles

- Mechanical waves have materials that are "waving" and can be described in terms of physical variables.
  - A string may be vibrating.
  - Sound waves are produced by molecules of a material vibrating.
  - Electromagnetic waves are associated with electric and magnetic fields.
- Waves associated with particles cannot be associated with a physical variable.

De-broglie hypothesis

De Broglie

In 1923 Prince Louis de-Broglie postulated that as nature exhibits a great amount of symmetry , wave-particle dualism need not be the special feature of light alone but the material particle must also exhibit such dual behaviour i.e. particle as well as wave nature.

The waves associated with moving particles are called matter waves or de Broglie waves, and the wavelength of matter waves are called de-Broglie wavelength.

de Broglie relation

$$\lambda = \frac{h}{p}$$

Planck's constant

de-Broglie wavelength

Momentum of particle

- **de-Broglie wavelength associated with accelerated charged particle:**

$$\lambda = \frac{h}{\sqrt{2meV}} = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m(K.E.)}} = \frac{h}{\sqrt{3mkT}}$$

For electron

$$\lambda = \frac{12.26}{\sqrt{V}}\, \mathrm{A}^{\mathrm{O}}$$

Properties of Matter Waves

- Wavelength of matter waves is given by , $\lambda = h/mv$.
- It is seen that as mass increases, its wavelength tends to Zero. Thus the wavelength of **macroscopic bodies** is **insignificant** in comparison to the size of body even at very low velocities.
- On the other hand, de-Broglie wavelength associated with **microscopic particle** is **significan**t .
- Matter waves are produced by the motion of particles and are independent of the charge. therefore they are neither electromagnetic nor acoustic waves but are new kind of waves.
- They can travel through vacuum and do not require any material medium for there propagation.
- Smaller the velocity of the particle. The longer is the wavelength of matter wave associated with it.
- The velocity of matter wave depends on the velocity of material particle and is not constant quantity.
- The velocity of matter wave is greater than the velocity of light.
- They exhibit diffraction phenomenon as any other waves.

WAVE NATURE OF PARTICLE AND DEBROGLIE HYPOTHESIS:

- We know that photon have both wave nature as well as particle nature.
- In 1924 French scientist Louis De Broglie found that photon have both natures, than particle must have both natures. In 1929 he was awarded a Noble prize on his discovery of wave nature of electron. He proposed that wave length associated with particle.
- $\lambda = \frac{h}{m_v}$ $\lambda = \frac{h}{\sqrt{2vemo}}$